

Literature Status for arXiv:1708.07333

Agent lane 8 note

2026-06-19

Status

The open norm-attainment-set question extracted from Sain's *On extreme contractions and the norm attainment set of a bounded linear operator* [1] is already answered in a later paper by Sain, *On the norm attainment set of a bounded linear operator and semi-inner-products in normed spaces* [2]. This packet is therefore classified as `literature_already_answered`, not as an original run result.

Source Question

In arXiv:1708.07333, Remark 2.1 on PDF page 3 says that, in view of the Hilbert-space theorem just proved there, for a bounded linear operator T between general Banach spaces X, Y , obtaining a complete characterization of M_T appears much more difficult and, to the author's knowledge, remains open.

Supporting Answer

The later arXiv:1802.10439 explicitly identifies the same problem. Its abstract states that it obtains a complete characterization of the norm attainment set of a bounded linear operator between any two real normed spaces, and says this answers the open question from Sain's earlier paper on the norm attainment set. The introduction restates the question: find a necessary and sufficient condition for $x \in S_X$ to satisfy $x \in M_T$, where T is a bounded linear operator defined on a Banach space X . It then says the result gives a complete solution without requiring completeness, finite dimensionality, or identical domain and range.

The decisive statement is Theorem 2.3 of [2]: if X, Y are real normed spaces, $T \in \mathcal{L}(X, Y)$, and $z \in S_X$, then $z \in M_T$ if and only if for every $x \in X$ there exist semi-inner-products $[\cdot, \cdot]_X$ and $[\cdot, \cdot]_Y$ on X and Y such that

$$[Tx, Tz]_Y = \|T\|^2[x, z]_X.$$

Scope

This answers the real-normed-space version of the source question. The source paper's ambient notation allows real or complex scalars, while the supporting Theorem 2.3 is stated for real normed spaces. A complex version should be checked separately before recording a full complex-scalar closure.

Evidence

The support is explicit rather than agent-inferred: the supporting abstract and introduction both say that the open question is answered, and Theorem 2.3 is the advertised characterization. Local run-index search found no existing packet for arXiv:1708.07333 or arXiv:1802.10439 before this packet was created.

References

- [1] D. Sain, *On extreme contractions and the norm attainment set of a bounded linear operator*, arXiv:1708.07333.
- [2] D. Sain, *On the norm attainment set of a bounded linear operator and semi-inner-products in normed spaces*, arXiv:1802.10439.